Basin MCMC Assignment 1

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The purpose of this assignment is to break down the development of the initial basin probability distribution and probability in the Basin MCMC algorithm.

Question 1. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Provide a function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_1$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_1 : \mathcal{X}_1 \to \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. a good approximation to the log posterior in the region of high probability around θ_0;
3. easy to compute defining quantities from θ_0;
4. easy to sample from;
5. as simple as possible.

Note that this function need only be specified up to an additive constant.

Solution. TODO: read over. We present the solution as follows:

1. Define f_1 as the second-order Taylor polynomial of the log posterior about θ_0;
2. Derive a formula for the gradient $g = \nabla \log p(\theta_0 \mid y)$;
3. Derive a formula for the negative Hessian $P = -\nabla^2 \log p(\theta_0 \mid y)$;
4. State the final formula for f_1.

1. For $\theta \in \mathcal{X}_1$, the log posterior is given by

$$\log p(\theta \mid y) = -\frac{1}{2\sigma^2} \|y - A(\theta)\|_2^2 + \log p(\theta) + C,$$

where C is some constant independent of θ, and $A : \mathcal{X} \to [0,1]^{N_t N_x}$ is the forward map. Performing a second-order Taylor expansion about θ_0 (since $\log p(\theta \mid y) \in C^3(U)$ for some open neighbourhood $U \subset \mathcal{X}_1$ of θ_0) gives

$$\log p(\theta \mid y) = \log p(\theta_0 \mid y) + g^\mathsf{T} (\theta - \theta_0) - \frac{1}{2} (\theta - \theta_0)^\mathsf{T} P (\theta - \theta_0) + O(\|\theta - \theta_0\|^3), \tag{1}$$

where

$$g := \nabla \log p(\theta_0 \mid y) \quad \text{and} \quad P := -\nabla^2 \log p(\theta_0 \mid y).$$

Motivated by (1), we define our approximation to the log posterior $f_1 : \mathcal{X}_1 \to \mathbb{R}$ by

$$f_1(\theta) := \log p(\theta_0 \mid y) + g^\mathsf{T} (\theta - \theta_0) - \frac{1}{2} (\theta - \theta_0)^\mathsf{T} P (\theta - \theta_0).$$

2. We have that

$$g = \nabla \log p(\theta_0 \mid y) = \nabla \left(\log p(y \mid \theta_0) + \log p(\theta_0) - \log p(y)\right) = \nabla \log p(y \mid \theta_0) + \nabla \log p(\theta_0). \tag{2}$$

For $\theta \in \mathcal{X}_1$, we have that

$$\nabla \log p(y \mid \theta) = \nabla \left(-\frac{1}{2\sigma^2}\|y - A(\theta)\|_2^2\right) = \frac{1}{\sigma^2} J_A(\theta)^\mathsf{T} \left(y - A(\theta)\right),$$

where $J_A(\theta)$ is the Jacobian of the forward map A (restricted to the domain $\mathcal{X}_1$). Recalling that $y = A(\theta_0)$,

$$\nabla \log p(y \mid \theta_0) = \frac{1}{\sigma^2} J_A(\theta_0)^\mathsf{T} \left(y - A(\theta_0)\right) = 0. \tag{3}$$

Finally, for $\theta \in \mathcal{X}_1$, we have that

$$\nabla \log p(\theta) = \nabla \log \left(\lambda_r \mathrm{e}^{-\lambda_r r}\right) = \nabla \left(-\lambda_r r\right) = (0, 0, -\lambda_r, 0)^\mathsf{T}. \tag{4}$$

Combining (2), (3), and (4), we have that

$$g = (0, 0, -\lambda_r, 0)^\mathsf{T}.$$

3. The negative Hessian of the log posterior is given by

$$P = -\nabla^2 \log p(\theta_0 \mid y) = -\nabla^2 \log p(y \mid \theta_0) - \nabla^2 \log p(\theta_0). \tag{5}$$

Recalling (4), we have that

$$\nabla^2 \log p(\theta_0) = 0. \tag{6}$$

Next, for $\theta \in \mathcal{X}_1$, we have that

$$\nabla^2 \log p(y \mid \theta) = -\frac{1}{\sigma^2} J_A(\theta)^\mathsf{T} J_A(\theta) + \frac{1}{\sigma^2} \sum_k (y - A(\theta))_k \nabla^2 A_k(\theta).$$

Again recalling that $y = A(\theta_0)$, we obtain

$$\nabla^2 \log p(y \mid \theta_0) = -\frac{1}{\sigma^2} J_A(\theta_0)^\mathsf{T} J_A(\theta_0). \tag{7}$$

Combining (5), (6), and (7), we have that

$$P = \frac{1}{\sigma^2} J_A(\theta_0)^\mathsf{T} J_A(\theta_0). \tag{8}$$

To compute P, we must compute $J_A(\theta_0)$. The forward map at (t, x) with parameters $\theta = (x_0, d, R, a)^\mathsf{T}$ is given by

$$A(t, x; \theta) = aB(x)w(t - T(x))$$

where

$$B(x) = \sqrt{\frac{d+R}{s(x)}} e^{-2\alpha s(x)}, \quad T(x) = \frac{2}{v}(s(x) - (d+R)) + \frac{2d}{v} = \frac{2}{v}(s(x) - R)$$

with $s(x) = \sqrt{(d+R)^2 + (x - x_0)^2}$, and

$$w(\tau) = (1 - 2(\pi f_0 \tau)^2) e^{-(\pi f_0 \tau)^2}.$$

We have that

$$\begin{aligned}
w'(\tau) &= 2(\pi f_0)^2 \tau (2(\pi f_0 \tau)^2 - 3) e^{-(\pi f_0 \tau)^2}, \\
\frac{\partial \log B(x)}{\partial x_0} &= (x - x_0) \left(\frac{1}{2s(x)^2} + \frac{2\alpha}{s(x)} \right), \\
\frac{\partial \log B(x)}{\partial (d+R)} &= \frac{1}{2(d+R)} - \frac{(d+R)}{2s(x)^2} - \frac{2\alpha(d+R)}{s(x)}, \\
-\frac{\partial T(x)}{\partial x_0} &= \frac{2(x-x_0)}{vs(x)}, \\
-\frac{\partial T(x)}{\partial d} &= -\frac{2(d+R)}{vs(x)}, \\
-\frac{\partial T(x)}{\partial R} &= \frac{2}{v} \left(1 - \frac{(d+R)}{s(x)} \right).
\end{aligned}$$

Applying the product and chain rules gives

$$\begin{aligned}
\frac{\partial A(t,x;\theta)}{\partial x_0} &= aB(x) \left[w(t - T(x)) \frac{\partial \log B(x)}{\partial x_0} + w'(t - T(x)) \frac{-\partial T(x)}{\partial x_0} \right], \\
\frac{\partial A(t,x;\theta)}{\partial d} &= aB(x) \left[w(t - T(x)) \frac{\partial \log B(x)}{\partial (d+R)} + w'(t - T(x)) \frac{-\partial T(x)}{\partial d} \right], \\
\frac{\partial A(t,x;\theta)}{\partial R} &= aB(x) \left[w(t - T(x)) \frac{\partial \log B(x)}{\partial (d+R)} + w'(t - T(x)) \frac{-\partial T(x)}{\partial R} \right], \\
\frac{\partial A(t,x;\theta)}{\partial a} &= B(x)\, w(t - T(x)).
\end{aligned}$$

Vectorising these four derivative images gives the four columns of J_0.

4. If P is positive definite, completing the square gives

$$\log p(\theta \mid y) \approx C - \frac{1}{2}(\theta - \mu)^\mathsf{T} P (\theta - \mu),$$

where

$$\mu = \theta_0 + P^{-1} g, \qquad \Sigma = P^{-1}.$$

Thus, after restricting it to the valid parameter domain, the local basin approximation is a truncated Gaussian with mean parameter μ and covariance parameter Σ.

In the implementation, the Moore–Penrose pseudoinverse P^+ is used in place of P^{-1} for numerical robustness when P is ill-conditioned:

$$\mu = \theta_0 + P^+ g, \qquad \Sigma = P^+.$$

If all eigenvalues of P are positive, this agrees with P^{-1}, up to the numerical tolerance used in computing the pseudoinverse. Small eigenvalues correspond to weakly identified directions and therefore large local posterior variances.

The local precision, covariance, and recentered mean then follow from

$$P = \frac{1}{\sigma^2} J_0^\mathsf{T} J_0, \qquad \Sigma = P^+, \qquad \mu = \theta_0 + \Sigma g.$$

Question 2. [pts.]

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top \in \mathcal{X}_1$. Provide a function that approximates the log likelihood $\log p(y \mid \theta)$ on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $g_2 : \mathcal{X}_2 \to \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable, i.e., $g_2(\theta_1, \theta_2) = g_2(\theta_2, \theta_1)$;
3. a good approximation to the log likelihood in the region of high probability around θ_0;
4. easy to compute defining quantities from θ_0;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant.

Solution. TODO: read over. Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\top \in \mathcal{X}_1$. We want to approximate the log posterior on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

For simplicity, we will start by approximating the log likelihood on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

This is given by

$$\log p(y \mid \theta_1, \theta_2) = -\frac{1}{2\sigma^2}\|A(\theta_0) - A(\theta_1) - A(\theta_2)\|_2^2 + C$$

for some constant C. Recall that the forward map is given by

$$A(t, x; \theta) = aB(x)w(t - T(x))$$

where

$$B(x) = \sqrt{\frac{d+R}{s(x)}}e^{-2\alpha s(x)}, \quad T(x) = \frac{2}{v}(s(x) - (d+R)) + \frac{2d}{v} = \frac{2}{v}(s(x) - R)$$

with $s(x) = \sqrt{(d+R)^2 + (x - x_0)^2}$, and

$$w(\tau) = (1 - 2(\pi f_0 \tau)^2)e^{-(\pi f_0 \tau)^2}.$$

Reparameterization Define $g_0 := (x_0, d_0, R_0)^\top$, so that we may write the representative atom as

$$\theta_0 = (g_0, a_0)^\top.$$

Now, with two atoms

$$\theta_1 = (g_1, a_1)^\top \quad \text{and} \quad \theta_2 = (g_2, a_2)^\top,$$

where $g_1 := (x_1, d_1, R_1)^\top$ and $g_2 := (x_2, d_2, R_2)^\top$, we will use the following reparameterization:

- $m := a_1 + a_2$ is the total amplitude,
- $s := \frac{a_1}{a_1 + a_2}$ is the amplitude split,

- $c := sg_1 + (1-s)g_2$ is the amplitude-weighted center,
- $h := g_1 - g_2$ is the difference.

We may recover the original parameters as follows:

- $a_1 = sm$,
- $a_2 = (1-s)m$,
- $g_1 = c + (1-s)h$,
- $g_2 = c - sh$.

Rewrite the forward map Define the function $F : [0, X_{\max}] \times [0, D_{\max}] \times [0, \infty) \to [0,1]^{N_t N_x}$ by

$$F(g) := A(g, 1),$$

where $A(g,1)$ is the $N_t N_x$-dimensional vector with $A(t, x; g, 1)$ as the $tN_x + x$-th entry. Notice that we have that

$$A(g, a) = aF(g).$$

Hence,

$$\begin{aligned} A(\theta_1) + A(\theta_2) &= a_1 F(g_1) + a_2 F(g_2) \\ &= m\left[sF(g_1) + (1-s)F(g_2)\right] \\ &= m\left[sF(c+\delta_1) + (1-s)F(c+\delta_2)\right]. \end{aligned} \tag{9}$$

where $\delta_1 := g_1 - c = (1-s)h$ and $\delta_2 := g_2 - c = -sh$.

Taylor approximation to F around c Now, (since F is smooth) performing a second-order Taylor expansion of F around c gives

$$F(c+\delta) = F(c) + J_c\delta + \frac{1}{2}H_c[\delta,\delta] + O(\|\delta\|^3), \tag{10}$$

where J_c is the Jacobian of F at c, given by

$$J_c = \left[\frac{\partial F(c)}{\partial x_0}, \frac{\partial F(c)}{\partial d}, \frac{\partial F(c)}{\partial R}\right]$$

and $H_c[\delta,\delta]$ is the $N_t N_x$-dimensional vector with k-th entry

$$\left(H_c[\delta,\delta]\right)_k = \delta^\mathsf{T}\nabla^2 F_k(c)\delta,$$

where $F_k(c)$ is the k-th entry of $F(c)$.

Using the Taylor approximation (10) for each term in (9) gives

$$\begin{aligned} sF(c+\delta_1) + (1-s)F(c+\delta_2) = F(c) + J_c\left(s\delta_1 + (1-s)\delta_2\right) + \frac{1}{2}\left(sH_c[\delta_1,\delta_1] + (1-s)H_c[\delta_2,\delta_2]\right) \\ + O(\|\delta_1\|^3 + \|\delta_2\|^3). \end{aligned} \tag{11}$$

Now, notice that since $s\delta_1 + (1-s)\delta_2 = 0$, we have that

$$J_c\left(s\delta_1 + (1-s)\delta_2\right) = 0. \tag{12}$$

Next, notice that since $H_c[\delta,\delta]$ is bilinear, we have that

$$sH_c[\delta_1,\delta_1] + (1-s)H_c[\delta_2,\delta_2] = s(1-s)^2 H_c[h,h] + (1-s)s^2 H_c[h,h]$$

$$= s(1-s)H_c[h,h]. \tag{13}$$

Combining (11), (12), and (13) gives

$$sF(c+\delta_1) + (1-s)F(c+\delta_2) = F(c) + \frac{1}{2}s(1-s)H_c[h,h] + O(\|\delta_1\|^3 + \|\delta_2\|^3). \tag{14}$$

Since

$$\|\delta_1\|^3 + \|\delta_2\|^3 = \left(|s|^3 + |1-s|^3\right)\|h\|^3,$$

if we restrict to the region where $|a_1 + a_2| > \epsilon$ for some $\epsilon > 0$ (which should be reasonable otherwise they cancel each other out in the image), so that $|s|, |1-s| < 1/\epsilon$, then we have that

$$O(\|\delta_1\|^3 + \|\delta_2\|^3) = O\left(\|h\|^3\right). \tag{15}$$

Overall, combining (14) and (15) gives

$$sF(c+\delta_1) + (1-s)F(c+\delta_2) = F(c) + \frac{1}{2}s(1-s)H_c[h,h] + O\left(\|h\|^3\right) \tag{16}$$

when $|a_1 + a_2| > \epsilon$.

Taylor approximation to F around g_0 Now, (since F is smooth) performing a first-order Taylor expansion of F around g_0 gives

$$F(g_0 + \delta) = F(g_0) + J_{g_0}\delta + O(\|\delta\|^2), \tag{17}$$

where and J_{g_0} is the Jacobian of F at g_0, given by

$$J_{g_0} = \left[\frac{\partial F(g_0)}{\partial x_0}, \frac{\partial F(g_0)}{\partial d}, \frac{\partial F(g_0)}{\partial R}\right].$$

Applying the approximation (17) to $F(c)$ in (16) gives

$$\begin{aligned} sF(c+\delta_1) + (1-s)F(c+\delta_2) &= F(c) + \frac{1}{2}s(1-s)H_c[h,h] + O\left(\|h\|^3\right) \\ &= F(g_0 + q) + \frac{1}{2}s(1-s)H_c[h,h] + O\left(\|h\|^3\right) \\ &= F(g_0) + J_{g_0}q + \frac{1}{2}s(1-s)H_c[h,h] + O\left(\|q\|^2 + \|h\|^3\right). \end{aligned} \tag{18}$$

when $|a_1 + a_2| > \epsilon$, where $q := c - g_0$.

Approximate $H_c[h,h]$ by $H_{g_0}[h,h]$ Since we would like an approximation that only depends on the representative atom (g_0, a_0), we will approximate $H_c[h,h]$. Given some assumptions (such as F has bounded third derivatives in a neighbourhood of g_0), we have that

$$H_c[h,h] = H_{g_0}[h,h] + O(\|q\|\|h\|^2).$$

Combining this with (18) gives

$$\begin{aligned} sF(c+\delta_1) + (1-s)F(c+\delta_2) = F(g_0) + J_{g_0}q + \frac{1}{2}s(1-s)H_{g_0}[h,h] \\ + O\left(\|q\|^2 + \|q\|\|h\|^2 + \|h\|^3\right), \end{aligned} \tag{19}$$

when $|a_1 + a_2| > \epsilon$.

Approximation of the forward map Define $\eta := m - a_0$ to be the difference between the total amplitude and the amplitude of the representative atom. Then, from (19) and (9) we have that

$$A(\theta_1) + A(\theta_2) = m\left[sF(c+\delta_1) + (1-s)F(c+\delta_2)\right]$$

$$
\begin{aligned}
&= (a_0 + \eta)\left[sF(c+\delta_1) + (1-s)F(c+\delta_2)\right] \\
&= (a_0 + \eta)\big[F(g_0) + J_{g_0} q + \frac{1}{2}s(1-s)H_{g_0}[h,h] \\
&+ O\left(\|q\|^2 + \|q\|\|h\|^2 + \|h\|^3\right)\big] \\
&= a_0 F(g_0) + \eta F(g_0) + a_0 J_{g_0} q + \frac{a_0}{2}s(1-s)H_{g_0}[h,h] \\
&+ O\left(|\eta|\,\|q\| + |\eta|\,\|h\|^2 + \|q\|^2 + \|q\|\|h\|^2 + \|h\|^3\right) \\
&= A(\theta_0) + \eta F(g_0) + a_0 J_{g_0} q + \frac{a_0}{2}s(1-s)H_{g_0}[h,h] \\
&+ O\left(|\eta|\,\|q\| + |\eta|\,\|h\|^2 + \|q\|^2 + \|q\|\|h\|^2 + \|h\|^3\right)
\end{aligned}
$$

when $|a_1 + a_2| > \epsilon$. That is,

$$
\begin{aligned}
A(\theta_1) + A(\theta_2) - A(\theta_0) &= \eta F(g_0) + a_0 J_{g_0} q + \frac{a_0}{2}s(1-s)H_{g_0}[h,h] \\
&+ O\left(|\eta|\,\|q\| + |\eta|\,\|h\|^2 + \|q\|^2 + \|q\|\|h\|^2 + \|h\|^3\right) \qquad (20)
\end{aligned}
$$

in some neighborhood around $\eta = m - a_0 = 0$.

Question 3. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Provide a function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_2$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_2 : \mathcal{X}_2 \to \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable, i.e., $f_2(\theta_1, \theta_2) = f_2(\theta_2, \theta_1)$;
3. a good approximation to the log posterior in the region of high probability around θ_0;
4. easy to compute defining quantities from θ_0;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant. This should build on the function found in Question 2.

Solution.

Question 4. **[pts.]**

Fix $n \geq 2$. Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\intercal \in \mathcal{X}_1$. Provide a function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_n$, where the observed data is the noise-free data $y = A(\theta_0)$ generated by this atom.

The function $f_n : \mathcal{X}_n \to \mathbb{R}$ should satisfy the following properties:

1. unimodal;
2. exchangeable;
3. a good approximation to the log posterior in the region of high probability around θ_0;
4. easy to compute defining quantities from θ_0;
5. easy to sample from;
6. as simple as possible.

Note that this function need only be specified up to an additive constant, and this constant need not be common to all n. This should be a generalization of the function found in Question 3.

Solution.

Question 5. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Rewrite the function found in Question 1 in the same form as the functions found in Question 4.

If it is not possible to write the function found in Question 1 in the same form as the functions found in Question 4, explain why. Then, provide an alternative function that approximates the log posterior $\log p(\theta \mid y)$ on $\mathcal{X}_1$ that is in the same form as the functions found in Question 4.

Solution.

Question 6. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Provide an approximation of the sequence of probabilities $\{p_n\}_{n\in\mathbb{N}}$ where

$$p_n := P(\theta \in X_n \mid y) = \int_{\mathcal{X}_n} p(\theta \mid y)\mathrm{d}\theta,$$

where $y = A(\theta_0)$ is the noise-free data generated by θ_0.

You may find it useful to use the functions found in Questions 4 and 5. Note that you need only approximate the sequence up to a common multiplicative constant; that is, it suffices to approximate quantities proportional to $\{p_n\}_{n\in\mathbb{N}}$.

Solution.

Question 7. **[pts.]**

Suppose we have a representative atom $\theta_0 = (x_{00}, d_0, R_0, a_0)^\mathsf{T} \in \mathcal{X}_1$. Provide an approximation of the probability p where

$$p :=,$$

where $y = A(\theta_0)$ is the noise-free data generated by θ_0.

Solution.